

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	32.9	12.4	2.2	0.8	0	51.7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	35.3	9	1.9	0.4	0.1	53.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	34.2	10.8	2.2	0.9	0.1	51.8
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	34.2	11	1.8	0.3	0	52.7
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	34.4	11.8	2.3	0.8	0.2	50.5
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	34.8	10.1	2.2	0.4	0	52.5
U2 - Efficacy additive alpha prior Beta(1, 1)	33.6	11.7	2.6	0.9	0.1	51.1
U3 - Efficacy additive alpha prior Beta(1, 1)	34.1	11.2	2.2	0.1	0	52.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.7	6	2.2	1.2	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.1	5.8	2.1	1.1	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	14.7	6.1	2.2	1.2	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.1	5.8	2.1	1.2	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	14.7	6.1	2.3	1.2	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	15	5.8	2.1	1.2	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	14.7	6.1	2.2	1.2	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	15.1	5.7	2.1	1.1	0.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.3	2.8	0.5	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.3	2.8	0.5	0.3	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.3	2.9	0.5	0.3	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.3	2.8	0.5	0.3	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.3	2.9	0.5	0.3	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.3	2.8	0.5	0.3	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	1.3	2.9	0.5	0.3	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	1.2	2.8	0.5	0.3	0.1	
Scenario 2						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.2	0.25	0.5	0.5	0.5	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	29.3	22.7	34.9	10.8	1.4	0.9
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	32.3	24.2	32.9	8.8	1	0.8
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	28.3	22.8	35.9	10.4	1.5	1.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	32.5	24.6	32.7	8.6	0.7	0.9
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	28.4	24.1	35.4	10.4	1.1	0.6
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	31.7	24	33.6	8.8	1.1	0.8
U2 - Efficacy additive alpha prior Beta(1, 1)	26.6	24.4	35	12	1	1
U3 - Efficacy additive alpha prior Beta(1, 1)	31.5	25.4	32.7	8.6	1	0.8
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.8	8	8.4	3.1	0.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.2	8.5	8.1	2.7	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.7	8.3	8.4	3	0.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.2	8.5	8.2	2.7	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.8	8.2	8.4	2.9	0.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.2	8.6	8.1	2.7	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.6	8.2	8.5	3.1	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	10.2	8.4	8.2	2.7	0.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	3.4	1.8	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	3.5	1.7	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	3.4	1.8	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	3.5	1.8	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	3.5	1.8	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	3.5	1.7	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.5	1.8	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	3.5	1.7	0.7	0.1	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 3						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.1	0.4	0.4	0.4	0.4	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.4	49.9	25.1	8.3	1.7	2.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.4	52.8	22.4	7.5	0.5	2.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.4	49.1	26	8.5	1.2	2.8
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.4	52.3	22	7.6	0.4	2.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.1	50	24.8	8.2	1.1	2.8
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	14.6	53.4	22	6.9	0.6	2.5
U2 - Efficacy additive alpha prior Beta(1, 1)	12.8	50.2	24.8	8.4	1.2	2.6
U3 - Efficacy additive alpha prior Beta(1, 1)	14.7	52.9	22.8	6.7	0.5	2.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.4	11.9	6.8	2.9	0.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.7	12.6	6.4	2.7	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.5	11.6	6.8	3.2	0.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.7	12.4	6.5	2.7	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.4	11.8	6.8	3.1	0.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.6	12.6	6.4	2.6	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	7.4	11.9	6.7	3	0.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	7.7	12.6	6.4	2.7	0.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	4	1.7	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	4	1.6	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	4	1.7	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	4.1	1.6	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	4.1	1.7	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	4.1	1.6	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	4	1.7	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	4.1	1.6	0.7	0.1	
Scenario 4						
DLT rate	0.05	0.1	0.18	0.3	0.4	
Efficacy rate	0.05	0.1	0.25	0.5	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.3	9	20.1	48.1	14.3	0.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	13.4	10	18.6	46.5	11.3	0.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.4	9.1	20.5	46.8	15	0.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.7	10.5	18.7	46.2	11.7	0.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8	8.8	21.1	48.3	13.6	0.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.5	10.5	20	45.8	11	0.2
U2 - Efficacy additive alpha prior Beta(1, 1)	9.3	9.1	19.9	47.9	13.6	0.2
U3 - Efficacy additive alpha prior Beta(1, 1)	13.2	10.4	19.5	45.4	11.3	0.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.4	4.9	7	10.3	3.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.7	5	7	10.5	2.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.3	4.9	7	10.3	3.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.6	5	7	10.4	2.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.4	4.9	7	10.3	3.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.7	5	7	10.4	2.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.3	4.9	7.1	10.3	3.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.7	5.1	7	10.5	2.8	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	3	1.7	2.4	0.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3	1.7	2.3	0.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	3	1.7	2.4	0.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3	1.7	2.3	0.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	3	1.7	2.4	0.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3	1.7	2.3	0.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.3	3	1.7	2.4	0.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	3	1.7	2.3	0.7	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.3	
Efficacy rate	0.05	0.3	0.3	0.3	0.3	
Utility 2	0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.6	31.9	27.8	23.1	11.2	0.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.1	40	26.2	19	7.3	0.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.7	31.9	27.6	24.4	10	0.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.9	40.1	26.3	17.6	7.7	0.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.2	32.4	27.3	23.1	10.6	0.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.4	40	24.8	19.9	7.5	0.4
U2 - Efficacy additive alpha prior Beta(1, 1)	6	31.3	27.8	23.8	10.7	0.4
U3 - Efficacy additive alpha prior Beta(1, 1)	8	39.3	25.5	19.1	7.7	0.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4	7.2	7.3	7.3	4.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.2	8.1	7.3	7	3.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4	7.3	7.3	7.3	4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.1	8.1	7.4	7.1	3.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4	7.4	7.2	7.3	4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.2	8.2	7.3	6.9	3.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	4	7.3	7.2	7.3	4.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.2	8.2	7.3	7	3.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3.2	2	2.3	1.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3.3	2	2.2	0.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3.2	2	2.3	1.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3.3	2	2.1	0.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3.2	2	2.3	1.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3.3	2	2.1	0.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	3.2	2	2.3	1.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	3.3	2.1	2.2	0.9	
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.3	0.32	0.35	0.38	0.4	
Utility 2	0	0	0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	62.3	17.5	3.1	1.4	0.2	15.5
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	66.4	15.2	2	1	0	15.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	61.4	18.2	3.8	1.2	0.2	15.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	66.4	15.4	2.3	0.8	0	15.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	62.5	18.2	3	1	0.2	15.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	65.7	15.4	2.2	1	0.2	15.5
U2 - Efficacy additive alpha prior Beta(1, 1)	62.4	17.1	3.6	1.5	0	15.4
U3 - Efficacy additive alpha prior Beta(1, 1)	67.1	15.1	2	1	0	14.8
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	18	6.4	1.9	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.6	6.3	1.8	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.2	6.3	1.9	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.5	6.3	1.8	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.1	6.4	1.9	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.5	6.3	1.7	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	18.1	6.4	2	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	18.7	6.3	1.7	0.7	0.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	2.5	0.5	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	2.4	0.5	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	2.5	0.5	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	2.4	0.5	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	2.5	0.5	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	2.4	0.5	0.2	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	2.5	0.5	0.2	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.8	2.4	0.5	0.1	0	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 7						
DLT rate	0.16	0.2	0.26	0.31	0.36	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.3	53.3	22.1	8	1.2	1.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.2	55.6	20.6	5.1	0.5	1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	14.7	53.9	20.8	8	1.6	1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.5	57.2	18.9	5.4	0.9	1.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.6	53.3	22.4	7.8	1.9	1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.5	57.1	19.6	5.3	0.5	1
U2 - Efficacy additive alpha prior Beta(1, 1)	14.2	52.8	22.6	7.3	1.9	1.2
U3 - Efficacy additive alpha prior Beta(1, 1)	16.5	56.7	19.4	5.9	0.4	1.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.2	13.6	5.7	1.9	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.5	14.5	5.2	1.4	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.3	13.6	5.6	2	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.6	14.5	5	1.5	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.2	13.5	5.8	1.9	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.4	14.7	5	1.5	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	8.1	13.6	5.8	1.8	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	8.5	14.6	5.1	1.5	0.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.7	1.3	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.7	1.2	0.4	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.6	1.3	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.7	1.1	0.4	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3.7	1.3	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3.7	1.1	0.4	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	3.7	1.3	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	3.7	1.1	0.4	0	
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.1	0.15	0.2	0.3	0.4	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	45.4	10.5	3.6	0.9	0.4	39.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	47.7	10	2.9	0.8	0.2	38.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	46.5	10.3	3.5	0.8	0.4	38.5
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	46.4	11.3	2.6	0.8	0.2	38.7
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	45.1	10.9	3.7	0.9	0.4	39
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	46.5	10.3	2.5	0.9	0.3	39.5
U2 - Efficacy additive alpha prior Beta(1, 1)	45.9	10	3.5	0.7	0.5	39.4
U3 - Efficacy additive alpha prior Beta(1, 1)	46.5	11.4	2.2	0.6	0.2	39.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.1	5.9	1.9	0.8	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.4	5.8	1.8	0.8	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.1	5.9	1.9	0.9	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.3	5.8	1.8	0.8	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	16	5.9	1.9	0.9	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.4	5.7	1.8	0.8	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	16.2	5.8	1.9	0.8	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	16.3	5.9	1.7	0.8	0.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.1	2.5	0.4	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	2.5	0.4	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	2.6	0.4	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	2.5	0.4	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	2.5	0.4	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	2.5	0.4	0.2	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	1	2.5	0.4	0.2	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	1	2.5	0.4	0.2	0	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.2	0.25	0.5	
Utility 2	-0.1	-0.1	0	0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.9	19.4	22.1	13.4	20.5	4.7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	25.3	19.9	19.4	12.5	17.3	5.6
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.1	18.9	22	12.7	21.1	5.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	26.2	19.6	19	11.4	18.1	5.7
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.6	19.7	21	14	20.6	5.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	25.2	22.1	17.8	13.1	16.4	5.4
U2 - Efficacy additive alpha prior Beta(1, 1)	20	18.4	21	14.3	21.1	5.2
U3 - Efficacy additive alpha prior Beta(1, 1)	26.5	19.8	18.4	12.3	17.6	5.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.8	6.7	5.9	4.9	4.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.5	6.5	5.6	5	3.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.7	6.7	5.9	5	4.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.5	6.7	5.6	4.9	3.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.8	6.8	5.8	5	4.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.5	6.6	5.6	4.9	3.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	7.7	6.7	5.8	5	4.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	8.4	6.5	5.6	4.9	4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	3.7	1.6	1.6	0.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	3.7	1.5	1.6	0.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	3.7	1.6	1.7	0.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	3.7	1.5	1.6	0.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	3.7	1.6	1.7	0.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	3.7	1.5	1.6	0.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	3.7	1.6	1.7	0.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.8	3.6	1.5	1.6	0.7	
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.4	0.42	0.45	0.5	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.6	38.6	13.6	4.3	1.1	21.8
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	21.5	40.3	11.8	3.6	0.8	22
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.5	38.7	13.5	4.1	1.3	21.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	21.9	39.5	12.3	3.8	0.7	21.8
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.9	40.5	12.4	4.7	1.3	21.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.9	40.6	12.4	3.3	1	21.8
U2 - Efficacy additive alpha prior Beta(1, 1)	20.7	38	14.2	3.8	1.5	21.8
U3 - Efficacy additive alpha prior Beta(1, 1)	21	40.2	13	2.7	1.1	22
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.9	10.4	4.3	1.7	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.1	10.7	4	1.6	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.9	10.4	4.2	1.8	0.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.2	10.7	4	1.6	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11	10.4	4.2	1.7	0.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.1	10.7	4.1	1.6	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	10.9	10.4	4.3	1.7	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	11	10.7	4.1	1.6	0.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	3.9	1	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	3.9	0.9	0.4	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	3.9	0.9	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	3.9	0.9	0.4	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	3.9	1	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	3.9	0.9	0.4	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.9	1	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	3.9	0.9	0.4	0.1	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	66	23.9	3.8	0.1	0	6.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	69.3	20.7	3.2	0.1	0	6.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	65.4	23.9	3.9	0.2	0	6.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	68.1	22.3	2.9	0.1	0	6.6
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	66.1	22.9	4.1	0.1	0	6.8
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	68.3	21.7	2.9	0	0.1	7
U2 - Efficacy additive alpha prior Beta(1, 1)	66	22.9	4.5	0.2	0	6.4
U3 - Efficacy additive alpha prior Beta(1, 1)	69.1	21.6	2.4	0.1	0	6.8
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.7	8.2	2.9	1.1	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.2	8	2.7	1	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.6	8.2	2.9	1.1	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.2	8	2.7	1	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.6	8.2	2.9	1.1	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.2	7.9	2.7	1	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	16.6	8.3	2.9	1.1	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	17	8.1	2.7	1	0.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.2	3.5	0.8	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.3	3.4	0.8	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.2	3.5	0.8	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.3	3.4	0.8	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.2	3.5	0.8	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.2	3.4	0.8	0.3	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	1.2	3.5	0.8	0.3	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	1.3	3.5	0.8	0.3	0	
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.1	0.12	0.4	0.6	
Utility 2	0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.3	15.9	13.4	37.5	18.6	0.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.9	16.1	11	36.8	15.9	0.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.1	14.9	13.7	38.4	17.5	0.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.6	16.6	9.7	36.4	16.3	0.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	14.8	14.8	13.7	38.2	18.3	0.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.2	16.9	10.6	35.6	16.3	0.4
U2 - Efficacy additive alpha prior Beta(1, 1)	13.7	14.5	13.7	39.4	18.3	0.4
U3 - Efficacy additive alpha prior Beta(1, 1)	21.3	16	11.4	35.1	15.8	0.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.8	5.6	5.8	9	4.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.2	5.8	5.6	9	4.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.8	5.6	5.8	8.9	4.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.3	5.8	5.6	9	4.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.8	5.6	5.8	8.9	4.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.3	5.8	5.6	9	4.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.8	5.6	5.8	9	4.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.2	5.8	5.6	9.1	4.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.2	1.6	2.3	1.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.3	1.6	2.2	1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.3	1.6	2.3	1.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.2	1.6	2.3	1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3.3	1.7	2.3	1.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.3	1.6	2.2	1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	3.2	1.6	2.4	1.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.2	1.7	2.2	1	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 13						
DLT rate	0.21	0.29	0.3	0.39	0.46	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.2	0	0	0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	21.4	40.3	12.9	6.2	1.2	18
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	24.2	39.4	11.6	4.8	1	19
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	22.3	38.4	13.5	6.1	1.2	18.5
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	23.3	40.8	11.7	5.1	0.8	18.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	22.6	38.3	13.1	6.5	1.2	18.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	24.4	38.4	11.8	5.6	0.8	19
U2 - Efficacy additive alpha prior Beta(1, 1)	22.4	38.2	13.2	6.1	1.5	18.6
U3 - Efficacy additive alpha prior Beta(1, 1)	24.7	38.8	11.8	4.9	0.9	18.9
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.7	9.7	4.6	2.6	0.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.9	9.9	4.4	2.4	0.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.7	9.7	4.5	2.6	0.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.9	9.9	4.4	2.4	0.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.6	9.7	4.6	2.6	0.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.9	9.8	4.5	2.4	0.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	10.7	9.7	4.6	2.5	0.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	11.1	9.9	4.3	2.4	0.6	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	4.1	1.1	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	4.1	1.1	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	4.1	1.1	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	4	1.1	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	4	1.1	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	4.1	1	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	4.1	1.1	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.9	4.1	1	0.7	0.1	
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.3	0.4	
Utility 2	0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	22	19.8	18	26.6	12.8	0.8
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	31.3	22.7	12.9	23.3	8.7	1.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	23.3	19.6	18.2	26	12	0.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	30.7	21.4	14.2	23.1	9.4	1.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	22.1	19.9	18.4	26.1	12.7	0.8
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	30.9	21.7	14.5	23.4	8.3	1.2
U2 - Efficacy additive alpha prior Beta(1, 1)	21.9	20.7	18.7	25.7	12	1
U3 - Efficacy additive alpha prior Beta(1, 1)	30.9	21.6	13.5	23.5	9.6	0.9
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	6.4	6.6	6	7.1	3.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	7	6.6	5.7	7	3.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	6.4	6.5	6	7.1	3.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	7	6.7	5.7	6.9	3.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.3	6.7	6	7	3.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	7	6.6	5.6	7	3.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	6.3	6.6	6	7.1	3.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	7	6.7	5.7	6.9	3.6	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	3.7	1.7	2.1	0.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	3.7	1.7	2	0.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	3.7	1.7	2.1	0.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	3.7	1.7	2	0.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	3.7	1.7	2.1	0.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	3.7	1.7	2.1	0.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	3.7	1.7	2.1	0.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.9	3.7	1.7	2.1	0.8	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 15						
DLT rate	0.13	0.2	0.32	0.39	0.46	
Efficacy rate	0.1	0.1	0.1	0.1	0.1	
Utility 2	0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	48.7	32.6	12.1	3.8	0.3	2.5
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	56	28.8	10	2.1	0.1	3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	48.8	33.1	11.8	3.7	0.3	2.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	55.5	29.7	9.5	2.2	0.1	3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	47.6	32.8	13.4	3.4	0.5	2.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	52.6	32.3	10	2.4	0.2	2.5
U2 - Efficacy additive alpha prior Beta(1, 1)	48.4	32.8	12.7	3.7	0.4	2
U3 - Efficacy additive alpha prior Beta(1, 1)	55.4	30.2	9.7	1.8	0.2	2.7
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.3	8.9	5.5	3.7	1.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.2	8.5	5.3	3.6	1.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.3	8.9	5.6	3.7	1.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.2	8.5	5.2	3.6	1.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.2	9	5.6	3.7	1.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.3	8.5	5.2	3.5	1.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	10.4	8.8	5.6	3.7	1.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	11.2	8.5	5.3	3.6	1.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.2	4.3	1.7	1.3	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.3	4	1.7	1.3	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.2	4.2	1.8	1.4	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.3	4.1	1.7	1.3	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.2	4.2	1.8	1.4	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.3	4.1	1.6	1.3	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	1.2	4.2	1.8	1.3	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	1.3	4.1	1.7	1.3	0.3	
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	34.9	29.8	16.4	12.2	6.7	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	40.3	29.2	16.3	10.2	4	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	34	29	18.1	11.7	7.2	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	41.1	31	14.3	9.2	4.4	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	34.3	29.3	17.4	12.1	6.9	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	40.3	30.8	15.4	9.2	4.3	0
U2 - Efficacy additive alpha prior Beta(1, 1)	35.6	28.8	16.6	12.3	6.7	0
U3 - Efficacy additive alpha prior Beta(1, 1)	40.5	30.5	15	9.5	4.5	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.3	5.3	6.1	8.2	6.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.6	5.8	6.2	8	5.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.3	5.2	6.1	8.2	6.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.6	5.7	6.3	8	5.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.3	5.3	6	8.1	6.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.5	5.7	6.3	8	5.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.3	5.2	6	8.2	6.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.6	5.8	6.3	8	5.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.8	2.2	2.9	1.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.9	2.3	2.9	1.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.8	2.2	2.9	1.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	2.9	2.4	2.8	1.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.8	2.2	2.9	1.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.9	2.3	2.9	1.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.8	2.2	2.9	1.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.9	2.4	2.9	1.4	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.3	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.9	60.6	16.9	1.4	0	0.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	23.5	62.1	13.3	1	0	0.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	22.7	59.5	16.9	0.7	0.1	0.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	23.8	61.8	13.4	0.8	0.1	0.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.7	60.5	16.7	0.9	0.1	0.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	23.4	62	13.8	0.7	0	0.1
U2 - Efficacy additive alpha prior Beta(1, 1)	21.7	60	17.1	1	0	0.2
U3 - Efficacy additive alpha prior Beta(1, 1)	23.2	62.3	13.7	0.7	0	0.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.5	15	4.8	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.9	15.5	4.1	0.5	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.5	15	4.7	0.7	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.9	15.5	4.1	0.5	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.5	15	4.7	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10	15.4	4	0.5	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.4	14.9	4.9	0.7	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	10	15.4	4.1	0.5	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.9	1.2	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	4	1	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.9	1.2	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	4	1	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.9	1.2	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.9	1	0.2	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	4	1.2	0.2	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.9	1	0.2	0	
Scenario 18						
DLT rate	0.02	0.04	0.1	0.2	0.35	
Efficacy rate	0.05	0.1	0.15	0.2	0.5	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.6	11.1	10.7	18.9	47.7	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.8	13.5	10.7	16.2	41.8	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.1	10.6	9.6	18.9	48.8	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.5	13	10.1	16.3	43.1	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.9	11	10.6	18.2	48.3	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.1	13.7	10.8	15.5	41.9	0
U2 - Efficacy additive alpha prior Beta(1, 1)	12	11.4	10.4	18.9	47.3	0
U3 - Efficacy additive alpha prior Beta(1, 1)	17.9	13.2	11	15.8	42.1	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.4	4	5.2	7.2	10.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.6	4.3	5.3	7.2	9.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.4	4	5.2	7.1	10.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.6	4.2	5.2	7.2	9.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.5	4	5.2	7.2	10.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.6	4.3	5.3	7.2	9.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.4	4	5.2	7.2	10.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.6	4.3	5.2	7.2	9.7	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.7	1.6	2.5	1.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.8	1.6	2.5	1.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.7	1.6	2.5	1.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.8	1.6	2.5	1.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.7	1.6	2.5	1.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.8	1.6	2.5	1.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.7	1.6	2.5	1.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.8	1.6	2.5	1.6	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.6	0.77	
Efficacy rate	0.08	0.3	0.32	0.35	0.4	
Utility 2	0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	21.8	57.4	16.2	1.8	0.1	2.7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	24.1	58.1	13.8	1.1	0	2.9
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	21.7	58.5	14.8	2	0	3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	25.2	56.9	13.3	1.7	0.1	2.8
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.5	59.2	15.5	2	0.1	2.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	25	56.6	14.1	1.3	0	3
U2 - Efficacy additive alpha prior Beta(1, 1)	21.3	59.2	14.6	1.9	0.1	2.9
U3 - Efficacy additive alpha prior Beta(1, 1)	24.2	58.1	13.3	1.3	0.2	2.9
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.6	12.6	6.3	2	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.9	12.9	5.9	1.9	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.6	12.6	6.3	2.1	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.9	12.9	5.9	1.9	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.6	12.6	6.3	2.1	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9	12.8	6	1.9	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	8.6	12.6	6.2	2.1	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	8.9	12.8	6	1.9	0.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	4.7	1.6	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	4.7	1.5	0.7	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	4.7	1.6	0.7	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	4.7	1.5	0.6	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.9	4.8	1.6	0.7	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.9	4.7	1.6	0.6	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	4.7	1.6	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.9	4.7	1.5	0.6	0	
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.9	49.4	28.3	8.9	1	0.5
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.4	51.8	23.8	7.3	0.4	0.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.8	48.2	29.9	8.7	1	0.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.6	51.8	24.9	6.8	0.5	0.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.9	47.4	29.4	8.8	1	0.5
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.4	50.7	24.4	7.4	0.5	0.6
U2 - Efficacy additive alpha prior Beta(1, 1)	12.1	48.4	28.2	10	0.8	0.5
U3 - Efficacy additive alpha prior Beta(1, 1)	14.9	52.2	23.7	8.2	0.6	0.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.1	10.3	8.2	4.8	1.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.4	10.7	7.9	4.6	1.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.1	10.2	8.3	4.9	1.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.5	10.8	7.9	4.5	1.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.1	10.3	8.2	4.8	1.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.4	10.8	8	4.5	1.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	5	10.3	8.3	4.9	1.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.4	10.7	7.9	4.5	1.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.9	2.3	1.6	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.9	2.3	1.5	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.9	2.3	1.6	0.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	4	2.3	1.5	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.9	2.3	1.6	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.9	2.3	1.5	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.9	2.3	1.6	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	4	2.3	1.5	0.4	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.5	0.5	0.5	0.5	0.5	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	53.5	2.1	0	0	0	44.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	53	2.6	0	0	0	44.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	53.3	2.2	0	0	0	44.5
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	53.6	1.9	0.1	0	0	44.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	52.7	2.4	0.1	0	0	44.8
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	53.6	2.1	0.1	0	0	44.2
U2 - Efficacy additive alpha prior Beta(1, 1)	52.9	2.8	0	0	0	44.3
U3 - Efficacy additive alpha prior Beta(1, 1)	53.4	2.3	0.1	0	0	44.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.6	2.8	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.6	2.8	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.4	2.8	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.6	2.7	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.4	2.9	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.6	2.7	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	19.5	2.8	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	19.6	2.7	0.1	0	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	1.1	0	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	1.1	0	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	1.1	0	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	1	0	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	1.1	0	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	1	0	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	1.1	0	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.3	1	0	0	0	
Scenario 22						
DLT rate	0.07	0.27	0.5	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.5	0.5	
Utility 2	0	0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	24.7	58.5	13.5	1.6	0.1	1.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	29.9	53.7	12.6	2.1	0	1.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	26.1	57.2	13.3	1.7	0	1.7
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	29.2	54.9	12.4	1.9	0	1.6
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	26.3	56.7	13.1	2.2	0.1	1.6
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	28.6	55.1	12.8	1.7	0	1.8
U2 - Efficacy additive alpha prior Beta(1, 1)	27.1	57.1	12.8	1.4	0.1	1.5
U3 - Efficacy additive alpha prior Beta(1, 1)	28.7	53.9	13.5	2	0	1.9
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.2	12.8	5.7	2.1	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.7	12.7	5.4	1.9	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.2	12.9	5.7	2.1	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.7	12.9	5.3	1.9	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.3	12.8	5.6	2.1	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.7	12.8	5.4	1.9	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.4	12.7	5.7	2.1	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.7	12.8	5.4	1.9	0.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.2	4.7	1.6	0.6	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.2	4.7	1.6	0.6	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.1	4.8	1.6	0.6	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.2	4.7	1.6	0.5	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.1	4.7	1.6	0.6	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.2	4.7	1.6	0.5	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	1.2	4.8	1.6	0.6	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	1.2	4.7	1.6	0.5	0	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 23						
DLT rate	0.32	0.55	0.8	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.6	3.8	0	0	0	75.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.1	3.6	0.1	0	0	77.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.3	3.8	0	0	0	75.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.1	4.5	0	0	0	75.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.8	4.1	0.1	0	0	76
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.5	4.2	0	0	0	76.3
U2 - Efficacy additive alpha prior Beta(1, 1)	21.3	3.5	0	0	0	75.2
U3 - Efficacy additive alpha prior Beta(1, 1)	19.3	4.3	0.1	0	0	76.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.9	4.9	0.9	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	15	4.7	0.9	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	15	4.9	0.9	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.1	4.7	0.9	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	15	4.8	0.9	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	15	4.8	0.9	0.1	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	15	4.9	1	0.1	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	15.1	4.7	0.9	0.1	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	2.2	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	2.1	0.2	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	2.1	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	2.1	0.2	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	2.1	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	2.1	0.2	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	2.1	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.8	2.1	0.2	0	0	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.7	0.7	3.6	48.1	45.9	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	2.5	1.5	4.9	53.5	37.6	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.4	0.9	3.3	47.7	46.7	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	2.6	1.4	4	52.9	39.1	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.7	0.8	3.6	48.2	45.7	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	2.5	1.5	4.6	52.5	38.9	0
U2 - Efficacy additive alpha prior Beta(1, 1)	1.1	0.9	3.6	49	45.4	0
U3 - Efficacy additive alpha prior Beta(1, 1)	2.6	1.1	4.2	52	40.1	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.1	3.2	3.8	11.8	8.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.1	3.2	3.9	12.6	7.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.1	3.1	3.8	11.8	8.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.1	3.2	3.9	12.6	7.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.1	3.2	3.8	11.8	8.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.1	3.2	3.9	12.4	7.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.1	3.2	3.8	11.9	8.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.1	3.2	3.9	12.4	7.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0	2.4	1	2.4	1.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0	2.4	1.1	2.5	1.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0	2.4	1	2.5	1.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0	2.4	1.1	2.5	1.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0	2.4	1.1	2.4	1.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0	2.4	1.1	2.5	1.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	0	2.4	1.1	2.4	1.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	0	2.4	1.1	2.5	1.5	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.2	0.25	0.4	0.45	0.5	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	28.7	39.1	28.2	3.9	0	0.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	34.2	35.5	26.7	3.4	0.1	0.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	30.2	36.4	29.2	4	0	0.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	34.3	36.4	25.5	3.6	0.1	0.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	28	38.3	29.8	3.8	0	0.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	34.7	36.5	25.7	3	0	0.1
U2 - Efficacy additive alpha prior Beta(1, 1)	30.4	36.8	29.2	3.5	0	0.1
U3 - Efficacy additive alpha prior Beta(1, 1)	34.9	36.2	25.5	3.3	0	0.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.8	10.4	8.2	2.4	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.3	10.3	8.1	2.2	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.8	10.2	8.3	2.4	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.4	10.3	8	2.2	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.7	10.2	8.3	2.5	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.2	10.4	8.2	2.2	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	8.8	10.2	8.4	2.4	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.3	10.3	8.1	2.2	0.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	3.9	2	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	3.9	1.9	0.8	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	3.9	2	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	3.9	1.9	0.8	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	3.9	2	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.9	3.9	1.9	0.8	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	3.9	1.9	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.9	3.9	1.9	0.8	0	
Scenario 26						
DLT rate	0.15	0.22	0.3	0.62	0.8	
Efficacy rate	0.1	0.2	0.35	0.45	0.55	
Utility 2	-0.1	0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	27.7	37.1	26.3	5.2	0.1	3.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	32.7	36	23	4.7	0.1	3.5
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	28.8	37.4	25.3	4.7	0.2	3.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	33.7	34.9	22.6	5.4	0.1	3.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	28.1	36.9	25.6	5.6	0.2	3.6
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	32.2	37.5	22.5	4.6	0.1	3.1
U2 - Efficacy additive alpha prior Beta(1, 1)	28.3	36.8	26.1	5	0.2	3.6
U3 - Efficacy additive alpha prior Beta(1, 1)	34.4	36.2	22	4.2	0.1	3.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.3	9.7	7.6	2.9	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.9	9.7	7.3	2.6	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.3	9.7	7.5	2.9	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.9	9.7	7.2	2.7	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.3	9.7	7.4	3	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.8	9.7	7.2	2.8	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.4	9.7	7.5	2.9	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.9	9.5	7.3	2.8	0.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	4.2	1.6	1	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	4.1	1.6	1	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	4.1	1.6	1	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	4.1	1.6	1	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	4.2	1.6	1.1	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.9	4.1	1.5	1	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.9	4.2	1.6	1	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.9	4.1	1.5	1	0.1	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
Utility 2	0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.9	34.7	38.1	9.1	0.2	1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	23.7	32.6	35.1	7.3	0.2	1.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.6	33.1	37.9	9.1	0.4	0.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	22.7	33.9	35.1	7.2	0.2	0.9
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.9	33.2	37.3	8.9	0.5	1.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	23.7	32.3	35.6	6.9	0.5	1
U2 - Efficacy additive alpha prior Beta(1, 1)	18.3	33.8	37.8	8.8	0.4	0.9
U3 - Efficacy additive alpha prior Beta(1, 1)	24.1	34.1	33.1	7.8	0.1	0.8
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	6.9	8.9	9.7	3.8	0.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.5	9.1	9.5	3.5	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	7	8.9	9.6	3.7	0.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.4	9.1	9.5	3.5	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	7	8.9	9.6	3.8	0.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.4	9.1	9.6	3.5	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	7	8.9	9.7	3.7	0.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	7.4	9.2	9.5	3.5	0.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.9	2.1	1.2	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	3.9	2	1.1	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.8	2.1	1.2	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	3.9	2	1.1	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.8	2.1	1.3	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	3.9	2	1.1	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.9	2.1	1.2	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	3.9	2	1.1	0.1	
Scenario 28						
DLT rate	0.2	0.3	0.5	0.6	0.77	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	30.2	43.7	5.4	0.5	0	20.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	31.6	43.9	4.6	0.2	0	19.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	29.9	44.2	5	0.7	0	20.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	31.4	43.1	4.9	0.6	0	20
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	29.6	44.1	5.3	0.6	0	20.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	32	42	4.5	0.4	0	21.1
U2 - Efficacy additive alpha prior Beta(1, 1)	31	43.6	5.1	0.5	0	19.8
U3 - Efficacy additive alpha prior Beta(1, 1)	31.5	42.9	4.7	0.4	0	20.5
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.2	11.1	3.8	1	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.5	11.1	3.6	0.9	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.2	11.2	3.7	1.1	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.6	11.1	3.5	0.9	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.3	11.1	3.7	1.1	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.6	11.1	3.5	0.9	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	12.3	11.2	3.7	1	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	12.6	11.1	3.5	0.9	0.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	4.3	1	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.1	4.3	0.9	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.1	4.4	0.9	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.1	4.3	0.9	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.1	4.4	1	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.1	4.3	0.9	0.3	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	1.1	4.4	0.9	0.3	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	1.1	4.3	0.9	0.2	0	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 29						
DLT rate	0.19	0.3	0.56	0.74	0.86	
Efficacy rate	0.4	0.4	0.4	0.4	0.4	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	58.3	34.4	4.6	0	0	2.7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	61.3	32.9	3.2	0	0	2.6
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	59.2	33.1	5.1	0.1	0	2.5
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	61.4	31.6	4.5	0	0	2.5
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	59.3	33.3	4.8	0.1	0	2.5
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	60.7	33.1	3.7	0	0	2.5
U2 - Efficacy additive alpha prior Beta(1, 1)	58.1	33.9	5.1	0	0	2.9
U3 - Efficacy additive alpha prior Beta(1, 1)	61.8	31.7	4.1	0	0	2.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.2	9.8	2.2	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.7	9.7	1.8	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.2	9.8	2.2	0.4	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.8	9.7	1.8	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.3	9.7	2.1	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.7	9.8	1.8	0.2	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	17.1	9.8	2.2	0.4	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	17.8	9.7	1.7	0.2	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	3.3	0.8	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	3.2	0.6	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	3.3	0.7	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	3.2	0.6	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.9	3.3	0.7	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.9	3.2	0.6	0.1	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	3.3	0.8	0.1	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.9	3.2	0.6	0.1	0	
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.2	0.4	0.5	
Utility 2	0.1	0.1	0	0.1	0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.5	16.7	16.5	38	11.3	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	24.8	19	13.1	33	10.1	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.4	15.4	17.1	37.7	11.4	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	25.6	17.8	13.5	33	10.1	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.9	16	16.6	37.2	12.3	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	23.7	19.3	14.3	33.7	9	0
U2 - Efficacy additive alpha prior Beta(1, 1)	18	16.4	15.2	38.5	11.9	0
U3 - Efficacy additive alpha prior Beta(1, 1)	25.9	17.3	14	32.7	10.1	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.4	5.3	6.4	9.9	3.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.7	5.8	6.3	9.7	3.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.4	5.3	6.3	10	4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.7	5.8	6.4	9.8	3.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.4	5.4	6.4	9.9	3.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.7	5.8	6.3	9.8	3.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.4	5.4	6.4	9.8	4	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.7	5.7	6.4	9.8	3.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.9	2	2.3	1.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3	2	2.2	1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	2.9	1.9	2.3	1.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	3	2	2.2	1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	2.9	2	2.3	1.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3	2	2.2	1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.3	2.9	2	2.3	1.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	3	2	2.2	1	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 31						
DLT rate	0.04	0.06	0.1	0.2	0.6	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	2.6	34.5	32.6	25.9	4.4	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4	39.9	32.4	19.9	3.8	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	2.7	33.8	33	26	4.5	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.2	39.5	33.2	20.2	2.9	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	2.5	34.6	32.5	25.1	5.3	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.1	40.9	31.9	19.6	3.5	0
U2 - Efficacy additive alpha prior Beta(1, 1)	2.9	34.9	32.7	24.9	4.6	0
U3 - Efficacy additive alpha prior Beta(1, 1)	4.6	39.8	31.6	21	3	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.9	11.6	8.5	4.9	1.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4	12.9	8.4	4	0.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.9	11.6	8.4	4.9	1.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4	12.8	8.5	4	0.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.9	11.6	8.5	4.8	1.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4	13	8.2	4.1	0.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.9	11.8	8.4	4.8	1.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.1	13	8.1	4.2	0.6	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0	2.7	1.8	1.2	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.9	1.7	1	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.7	1.7	1.2	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.8	1.7	1	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.7	1.7	1.2	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.8	1.7	1	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.7	1.8	1.2	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.8	1.7	1	0.2	
Scenario 32						
DLT rate	0.02	0.05	0.11	0.3	0.36	
Efficacy rate	0.08	0.15	0.3	0.3	0.3	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	12	13.7	32.7	29.1	12.5	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.2	16.5	32.5	25.9	7.9	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.7	13.7	33.5	28.7	12.4	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.4	16.5	31.7	26.5	7.9	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.1	13.7	32.1	30.3	11.8	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.2	17.4	31.8	27.4	7.2	0
U2 - Efficacy additive alpha prior Beta(1, 1)	11.1	13.7	31.8	30.9	12.5	0
U3 - Efficacy additive alpha prior Beta(1, 1)	16.6	17.5	31.5	27.6	6.8	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.5	4.4	8.2	8.9	5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.7	4.7	8.5	9	4.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.5	4.4	8.3	8.9	4.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.7	4.9	8.4	8.8	4.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.5	4.4	8.3	9	4.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.6	4.8	8.5	8.9	4.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.5	4.4	8.2	9	4.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.7	4.8	8.4	9	4.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.7	1.9	2.9	1.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.8	2.1	2.8	1.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.7	2	2.9	1.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.8	2	2.8	1.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.7	1.9	3	1.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.8	2.1	2.8	1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.7	1.9	2.9	1.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.8	2	2.8	1.1	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.2	0.4	0.5	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.7	6.3	14.9	42.5	27.6	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.1	7.8	14.7	41.1	22.3	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.5	6.6	14.8	43.2	25.9	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	14.5	7.7	14.1	40.9	22.8	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.7	6.6	15.3	42.1	26.3	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	14.1	7.6	13.9	42.5	21.9	0
U2 - Efficacy additive alpha prior Beta(1, 1)	9.5	6.6	15.1	41.9	26.9	0
U3 - Efficacy additive alpha prior Beta(1, 1)	14.4	8.1	13.9	41.6	22	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.3	3.8	6.1	10.4	6.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.5	3.9	6.1	10.8	5.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.3	3.8	6.1	10.5	6.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.5	3.9	6	10.7	5.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.3	3.8	6.1	10.4	6.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.5	3.9	6.1	10.8	5.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.3	3.8	6	10.4	6.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.5	3.9	6.1	10.8	5.7	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.7	1.6	2.8	1.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.7	1.6	2.7	1.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.7	1.6	2.8	1.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.7	1.6	2.7	1.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.7	1.6	2.8	1.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.7	1.6	2.7	1.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.7	1.6	2.8	1.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.7	1.6	2.7	1.2	
Scenario 34						
DLT rate	0.05	0.13	0.3	0.36	0.42	
Efficacy rate	0.05	0.1	0.3	0.32	0.35	
Utility 2	0	0	0	0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.8	17.1	43.6	18.1	3.2	0.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	21.1	20.3	39.6	15.9	2.9	0.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.4	15.2	45.8	17.2	4.2	0.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	23.3	17.7	39.8	16.4	2.6	0.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.5	15.6	45.8	17.5	3.5	0.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	23.2	17.9	39.8	16.3	2.6	0.2
U2 - Efficacy additive alpha prior Beta(1, 1)	17.7	16.4	43.6	18.7	3.4	0.2
U3 - Efficacy additive alpha prior Beta(1, 1)	22.7	18.7	38.5	17.4	2.5	0.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.5	7	9.2	6.2	2.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	6	7	9.1	6.1	1.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.4	6.9	9.3	6.2	2.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	6	7	9.2	6.1	1.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.4	6.9	9.4	6.2	2.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.9	7	9.2	6	1.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	5.4	6.9	9.4	6.3	2.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	6.1	6.9	9.1	6.1	1.8	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.5	2.2	1.9	0.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	3.5	2.2	1.8	0.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.6	2.2	1.9	0.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	3.6	2.2	1.8	0.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.5	2.3	1.9	0.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	3.6	2.2	1.8	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.5	2.2	1.9	0.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.8	3.5	2.2	1.8	0.5	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 35						
DLT rate	0.03	0.09	0.3	0.4	0.45	
Efficacy rate	0.2	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.7	40.1	37.7	13	1.5	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.9	45.1	33.9	9.6	0.5	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.2	40.8	37.8	12.4	1.8	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.1	47.3	33.2	8.7	0.7	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.6	41.5	36.9	11.9	2.1	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.2	46.6	33.9	8.4	0.9	0
U2 - Efficacy additive alpha prior Beta(1, 1)	7.7	42.3	36.2	12.1	1.7	0
U3 - Efficacy additive alpha prior Beta(1, 1)	10.9	45	33.8	9.3	1	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.5	12	9.1	3.6	0.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.6	13.1	8.8	3.1	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.5	12	9.1	3.7	0.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.7	13.3	8.7	2.9	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.5	12	9.1	3.6	0.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.7	13.2	8.7	2.9	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.4	12.3	9	3.6	0.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.6	13.2	8.7	3	0.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3.1	2.2	1.1	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3.2	2.1	0.9	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3.1	2.2	1.1	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3.2	2.1	0.9	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3.1	2.1	1.1	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3.2	2.1	0.9	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	3.1	2.2	1.1	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	3.2	2.2	0.9	0.1	
Scenario 36						
DLT rate	0.13	0.29	0.35	0.4	0.43	
Efficacy rate	0.05	0.2	0.22	0.3	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	25.9	44.2	15.8	4.8	2.3	7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	31.5	41.3	14	4.7	1.7	6.8
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	26.5	44	15.6	4.9	2.4	6.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	30.1	42.9	13.1	5	1.6	7.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	25.1	43.9	16.8	5.1	2.2	6.9
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	31	43.4	13	4.7	1.7	6.2
U2 - Efficacy additive alpha prior Beta(1, 1)	26.6	41.3	17	5.8	2.5	6.8
U3 - Efficacy additive alpha prior Beta(1, 1)	29.1	42.9	14.3	5.3	1.6	6.8
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.4	10.3	5.4	3.3	1.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.8	10.3	5.3	3.1	1.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.4	10.3	5.4	3.2	1.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.8	10.2	5.3	3.2	1.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.4	10.3	5.4	3.3	1.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.9	10.3	5.3	3.2	1	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.4	10.3	5.4	3.2	1.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.8	10.2	5.4	3.1	1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	4.6	1.4	1	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	4.5	1.4	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	4.5	1.4	1	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	4.5	1.4	1	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.9	4.5	1.4	1	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	4.5	1.4	1	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.9	4.5	1.4	1	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	1	4.5	1.4	1	0.2	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 37						
DLT rate	0.08	0.3	0.4	0.45	0.5	
Efficacy rate	0.05	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	26.3	54.1	13.9	2.6	0.6	2.5
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	32.3	51	10.3	3.6	0.2	2.6
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	25.6	53.9	13.8	3.1	0.7	2.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	32.7	49.5	11.8	2.7	0.3	3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	28.2	52.2	13.9	2.4	0.7	2.6
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	33.6	49.7	10.8	2.8	0.5	2.6
U2 - Efficacy additive alpha prior Beta(1, 1)	24.9	54.8	14	3.5	0.5	2.3
U3 - Efficacy additive alpha prior Beta(1, 1)	31.1	51.4	12	2.5	0.3	2.7
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.9	11.4	5.8	2.9	0.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.5	11.4	5.5	2.7	0.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.9	11.4	5.7	2.9	0.9	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.5	11.3	5.5	2.8	0.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.9	11.5	5.8	2.8	0.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.4	11.4	5.5	2.8	0.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	9	11.5	5.7	2.8	0.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.5	11.3	5.5	2.8	0.8	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	4.9	1.6	1	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.1	4.8	1.5	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	4.9	1.6	1	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.1	4.8	1.5	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	4.9	1.6	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.1	4.8	1.5	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	1	4.9	1.6	1	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	1.1	4.8	1.5	0.9	0.2	